ABDULLAH AL OTAIBI

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EDUCATION

The Pennsylvania State University, University Park — Expected Graduation: December 2026

Bachelor of Science in Mechanical Engineering

Major GPA: 4.00 / 4.00

Relevant Coursework: Mechatronics (ME 454), Vibration of Mechanical Systems (ME 370), Design Methodology (ME 340), Product Design & Manufacturing Processes (IE 312), Thermodynamics, Fluid Mechanics, Heat Transfer

HONORS & AWARDS

- **Aramco CCP Scholarship Recipient** (Oct 2024 – Present) — Awarded through Saudi Aramco's highly competitive College Continuation Program to top Saudi students worldwide; full undergraduate sponsorship with guaranteed engineering placement upon graduation.
- **Tau Beta Pi Engineering Honor Society** — Initiated Spring 2026
- **President Walker Award** — Awarded for cumulative GPA of 4.00 in 12–36 graded Penn State credits

ENGINEERING PROJECTS

Nittany Lion Light Stand — Arduino Desk Lamp — Personal Project

- Designed and 3D-printed a lion-shaped lamp housing; integrated an Arduino-controlled servo arm for dynamic repositioning of the light source.
- Programmed servo motion sequences and LED control logic in Arduino C++; assembled full electromechanical system from hardware selection through final calibration.

Autonomous Wall-Following Robot — ME 454 Mechatronics

- Implemented PD and finite-state-machine controllers in Arduino C++ on a RedBot platform using IR sensor feedback for closed-loop wall tracking.
- Tuned controller gains across multiple hardware units; diagnosed and resolved motor wiring asymmetry through systematic debugging.

Pendulum Tuned Mass Damper Design — ME 370 Vibration of Mechanical Systems

- Derived 4-DOF nonlinear equations of motion via Euler-Lagrange formulation for a three-story shear building with a PTMD; linearized via small-angle approximation.
- Implemented modal superposition in MATLAB and ran a parametric grid search to identify optimal PTMD parameters meeting a 7 cm peak-displacement constraint under impulse loading.

e-Formula Car Body Aerodynamic Optimization — ME 340 Design Methodology

- Competed in a semester-long Penn State e-Formula design competition against 64 teams, with the goal of achieving the fastest car through the application of mechanical engineering principles including aerodynamics, drag reduction, lift optimization, and vehicle dynamics.
- Developed three original car body designs beyond the baseline in SolidWorks, iteratively refining geometry to reduce aerodynamic drag and improve downforce at sustained high-speed lap conditions.
- Validated performance through wind tunnel testing and CFD analysis; correlated experimental and simulated results to projected lap-time improvements, advancing to the quarterfinals.

PROFESSIONAL EXPERIENCE

AI-Driven Materials & Manufacturing Program — May 2026 – June 2026

- Selected for Prof. Hongtao Sun's competitive undergraduate research program at Penn State, focused on applying machine learning to advanced manufacturing and materials science.
- Completed Phase 1 training covering supervised learning, neural networks, and data-driven modeling techniques.

Penn State Department of Economics | Economics — Aug 2024 – Dec 2024

Grader

- Graded homework and exams for ECON 102 under Prof. Dave Brown, ensuring fairness and consistency in evaluation.
- Provided written feedback to students and proctored exams alongside the instructor.

LEADERSHIP & ACTIVITIES

Penn State Table Tennis Club | Secretary & Event Planner — Aug 2024 – Feb 2025

- Coordinated with Penn State Competitive Sports staff to organize community events and recruitment drives.
- Promoted the club at the Fall 2024 Involvement Fair and represented Penn State at regional tournaments.

SKILLS

Software & Programming: SolidWorks, MATLAB / Simulink, Arduino IDE, C / C++, Microsoft Excel

Engineering Methods: Closed-loop control (PD, PID, FSM), embedded systems, CFD, modal analysis

Languages: English (fluent), Arabic (native)